

2023 Spring CH223 Organic Chemistry II. Topic list for the 3rd mid-term exam.

The concepts, reaction conditions (e.g. reagents), reaction product structure, and arrow-pushing reaction mechanisms of the following topics should be mastered for the exam (unless otherwise noted). See the exam practice problem to see how those will be asked in the exam.

The topics of the previous exams will not be the main focus of this exam, but some concepts/terms/reactions of the previous topics may be involved for this exam (e.g. benzene is the 6-member ring aromatic compound which was learned in the exam1 topics).

A topic highlighted in cyan is about reaction mechanisms. The cyan highlight does NOT indicate that it is more important than other topics.

Class24 Carboxylic acid derivatives intro (Loudon 21, Carey 20)

Esters are more reactive than amides like aldehydes are more reactive than ketones (see the blackboard image)

Base hydrolysis of esters (saponification) converts esters into carboxylic acid by kicking out alcohol.

Base hydrolysis of esters mechanism

Class25 Carboxylic acid derivatives hydrolysis (Loudon 21, Carey 20)

Acid-catalyzed hydrolysis of esters also converts esters into carboxylic acid by kicking out alcohol like the base hydrolysis (saponification).

Acid-catalyzed hydrolysis of esters mechanism

Acid or base hydrolysis of amides converts amides into carboxylic acid by kicking out amines

Acid and base hydrolysis of amides are the fundamentally identical to the ester hydrolysis

Acid or base hydrolysis of nitriles converts nitriles into carboxylic acid by kicking out ammonia (NH₃)

Reactions of acid chlorides with amines produce amides.

Mechanism for "reactions of acid chlorides with amines produce amides"

Reactions of acid chlorides with alcohols produce esters.

Reactions of acid chlorides with carboxylic acids produce anhydrides.

Class26 Carboxylic acid derivatives nucleophiles and reduction (Loudon 21, Carey 20)

Reactions of anhydrides are similar to those of acid chloride: Anhydrides+amines producing amides and anhydrides+alcohol producing esters

Reduction of esters with lithium aluminum hydride (but not NaBH₄) produces primary alcohols.

Mechanism for "Reduction of esters with lithium aluminum hydride (but not NaBH₄) produces primary alcohols"

Reduction of amides with lithium aluminum hydride (but not NaBH₄) produces amines.

Mechanism for "Reduction of tertiary amides with lithium aluminum hydride (but not NaBH₄) produces amines." Reduction mechanism for primary and secondary amides do not have to be memorized.

Class27 Carboxylic acid derivatives Grignard and summary (Loudon 21, Carey 20)

Reduction of nitriles with lithium aluminum hydride (but not NaBH₄) produces primary amines.

Reaction of esters with Grignard reagents produces tertiary alcohols.

Mechanisms for nitrile reduction and ester+Grignard don't need to be memorized.

Class28 Enolate introduction (Loudon 22, Carey 21)

Enolate is a nucleophilic species derived from a carbonyl group under basic conditions

Enol (basically like protonated enolate) is a nucleophilic group derived from a carbonyl group under acidic conditions

Alpha proton (alpha position is next to carbonyl carbon) is unusually more acidic than regular C-H protons. This makes the alpha position become negatively charged/nucleophilic.

Alpha proton (H) can be replaced with deuterium (D) under acidic or basic conditions

Acid-catalyzed bromination of alpha position produces alpha-(mono)bromo product.

Mechanism for "acid-catalyzed bromination of alpha position produces alpha-(mono)bromo product."

Base bromination of alpha position produces alpha-tribromo product. If the original alpha position is CH₃, the base bromination produces carboxylic acid through the haloform reaction (like bromoform or chloroform).

Mechanism for Base bromination of alpha position. Both tribromo formation and haloform reaction.

Bromination of alpha position of carboxylic acid requires PBr₃ (HVZ reaction). Mechanism doesn't have to be memorized.

Class29 Enolate aldol reaction (Loudon 22, Carey 21)

Reaction of two aldehyde molecules (or two ketones or aldehydes/ketones) are called aldol reaction.

Base-catalyzed aldol reaction produces beta-hydroxyl carbonyl products.

Base-catalyzed aldol reaction mechanism.

Base aldol condensation (under heated conditions produces) produces alpha-beta unsaturated carbonyl

Base-catalyzed aldol reaction mechanism (which is base aldol reaction + dehydration).

Acid-catalyzed aldol condensation produces alpha-beta unsaturated carbonyl.

Acid-catalyzed aldol reaction mechanism (which is acid aldol reaction + dehydration).

Cross-aldol is aldol reaction/condensation between two different aldehydes ketones.
Intramolecular aldol reaction/condensation occurs in the same molecule.

Class30 Malonate chemistry (Loudon 22, Carey 21)

Claisen condensation is like aldol reaction but with esters (not aldehydes nor ketones) producing beta-keto ester

Claisen condensation mechanism (unlike aldol reactions, ester has a leaving group on carbonyl, which is OR, and ketone will be produced).

Intramolecular Claisen condensation is called Dieckmann condensation.

Claisen condensation of ester with aldehyde or ketone is possible (cross Claisen condensation).

Malonic ester (or malonate) is easier to generate reactive enolate or enol because of the lower pKa of the alpha C-H.

Alkylation of alpha position of malonates with alkyl bromide.

Mechanism for "Alkylation of alpha position of malonates with alkyl bromide."

Knoevenagel condensation is aldol condensation of malonate with aldehyde.

Mechanism of Knoevenagel condensation does not have to be memorized.

(Decarboxylation can be used after Claisen condensation after ester hydrolysis (this basically removes the whole ester group).)

Class31 Michael addition (Loudon 22, Carey 21)

If there is a C=C bond attached to carbonyl (called alpha, beta-unsaturated carbonyl compounds), the beta position is electrophilic.

Addition of nucleophile to the beta position of alpha, beta-unsaturated carbonyl is called conjugate addition (or Michael addition or 1,4-addition).

Conjugate addition with a cyanide reagent produces beta-nitrile carbonyl compound.

Mechanism of conjugate addition with a cyanide reagent

Conjugate addition with an amine reagent produces beta-amino carbonyl compound.

Conjugate addition of enolate/enol.

Mechanism of conjugate addition with enolate/enol.

Typical hydride reagents (NaBH_4 and LiAlH_4) causes direct addition (1.2-addition or carbonyl addition) rather than conjugate addition.

Organolithium reagents (like a carbanion) causes direct addition (1.2-addition or carbonyl addition) rather than conjugate addition. But, if organocuprate reagents (copper) are used, conjugate addition will be preferred than the 1.2-addition or carbonyl addition.